MINISTRY OF EDUCATION AND TRAINING

FPT UNIVERSITY — FPTU QUY NHON AI CAMPUS

FU CAHATA · FACULTY OF INFORMATION TECHNOLOGY — SOFTWARE ENGINEERING

OMNICAST | BROADCAST INTELLIGENCE PLATFORM

FINAL PROJECT DELIVERABLE REPORT

COURSE: PRN232 — BUILDING CROSS-PLATFORM BACK-END APPLICATION

OMNICAST: DIGITAL BROADCAST SCHEDULING & DELIVERY SYSTEM

Broadcast. Connect. Discover.

NestJS and gRPC microservice; Next.js 15 web portal; real-time LiveStream and Live Studio; EPG scheduling; AI Curator

Mentor / Supervisor:	**Mr. Huỳnh Vang** (Software Engineering Dept. — FPTU)
Campus & Team Note:	**FPTU Quy Nhon AI Campus — FU Cahata**
Academic Semester:	Fall 2026 (FA26)
Project Team Members (4):	**1. Nguyễn Bình An** (Team Leader / Backend & Security)
	2. Trần Thanh Sơn — MSSV: `QE170080` (gRPC Audit & Database)
	3. Nguyễn Vĩnh Phú — MSSV: `QE180136` (Mobile Flutter Lead)
	4. Đồng Hoàng Nguyên (Full-Stack Web, Live & AI)
GitHub Repository:	https://github.com/TomOutfit/OmniCast_PRN232-PRM393_Project_Fa26Semester
Live Deployment:	https://omnicast-fe.vercel.app (API: /api/v1)
Scale & Verification:	**142,897 LOC | 468/468 tests in archived CI run**

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Building a Cross-Platform Back-End Application: The OmniCast Digital Broadcast Scheduling and Delivery System

Nguyễn Bình An, **Trần Thanh Sơn** (QE170080), **Nguyễn Vĩnh Phú** (QE180136), **Đồng Hoàng Nguyên**

Mentor / Supervisor: **Mr. Huỳnh Vang**
FPTU Quy Nhon AI Campus — FU Cahata, Faculty of Information Technology, FPT University
Integrated PRN232/PRM393 project report

***Abstract*— OmniCast is a multi-platform digital broadcast scheduling, real-time LiveStream and content delivery system developed at FPTU Quy Nhon AI Campus — FU Cahata for the PRN232 course under the supervision of Mentor Mr. Huỳnh Vang. It comprises a NestJS back end with 19 business modules and 140+ REST endpoints, a Next.js 15 web portal with 33 routes (including a 7-theme interactive LiveStream room, a 4-mode Live Studio console and a full administrative suite), a 31-screen Flutter mobile client, and a gRPC audit-logging microservice. Four principal architectural contributions are presented: (1) an end-to-end bi-directional LiveStream architecture combining WebRTC WHIP/WHEP, MediaMTX HLS and a Socket.IO `/live` gateway supporting live chat, floating reactions, 6-tier Super Chat donations, pinned broadcaster announcements and zero-downtime on-the-fly program switching; (2) a broadcast-schedule conflict-detection algorithm implemented as a single raw SQL query that computes the effective end time of an existing event; (3) a 24/7 EPG filler whose seeded selection guarantees that every channel-day grid is both complete and reproducible; and (4) an AI curation pipeline that issues a single large language model call, validates the output against a Zod schema, offers a heuristic fallback path and records full version provenance. The database schema comprises 21 Prisma models on PostgreSQL. The complete system has been deployed on Vercel and validated by 468 automated tests in archived CI run 40630318005: 163 backend, 228 web and 77 mobile tests, all of which passed.**

***Index Terms*—broadcast scheduling, EPG, LiveStream, WebRTC WHIP/WHEP, Socket.IO, NestJS, Prisma, Next.js, gRPC, RBAC, Flutter, AI curation, conflict detection, FPTU Quy Nhon AI Campus, FU Cahata.**

I. Introduction

In recent years, the way Vietnamese audiences consume television and interactive live broadcasts (LiveStream) has changed dramatically. Viewers no longer sit in front of a screen at a fixed broadcast slot; they browse on-demand, join interactive live rooms with real-time chat and floating reactions, and expect content recommendations tailored to their interests. Broadcasters and digital-content providers, in turn, need a single, reliable and scalable scheduling and Live Studio control system that serves several delivery surfaces at once: the web, mobile applications and live-streaming devices.

This paper presents **OmniCast** — a multi-platform digital broadcast scheduling, real-time LiveStream and content delivery system developed at **FPTU Quy Nhon AI Campus — FU Cahata** for the course **PRN232 — Building Cross-Platform Back-End Application** under the academic mentorship of **Mr. Huỳnh Vang**. The system comprises a high-performance NestJS back end that serves both a Next.js 15 web portal (33 routes) and a Flutter mobile application (31 screens), with the core business logic concentrated on four capabilities: a bi-directional real-time LiveStream and Live Studio ecosystem, an electronic programme guide (EPG) that is never empty, an atomic broadcast-schedule conflict-detection mechanism, and an AI editorial assistant with content-compliance validation.

A. Motivation and Problem Statement

Four technical problems dominate the system's design, each derived from concrete project requirements:

1) **Multi-source LiveStream ingestion and real-time studio orchestration.** A modern broadcast platform must allow producers to go live from curated VOD libraries, device cameras/screens over WebRTC WHIP, or verified public catalogs, while supporting zero-downtime on-the-fly program switching without dropping viewers who are actively chatting, sending floating reactions and donating via Super Chat.
2) **Sparse schedules.** A real broadcast channel originates only a handful of productions per day. Building a 24-hour EPG grid directly from the event table leaves most of the grid empty and renders the day filter in the user interface meaningless. The system therefore requires an automatic filler that guarantees every channel-day grid is fully populated with valid content.
3) **Schedule conflicts.** If two programmes on the same channel overlap, viewers cannot tell which one is actually on air. The validation must detect all three overlap forms (head, tail, full containment) and must execute as a single database query to avoid race conditions.
4) **Manual content moderation.** The editorial workflow (writing descriptions, tagging, categorising) is manual and time-consuming. Automating it with a large language model reduces the load, but requires a fallback path when

no API key is present, and the output must be schema-validated before it is persisted.

B. Contributions

This paper makes the following technical contributions:

1) A layered **NestJS + Prisma + PostgreSQL** architecture with 19 business modules, 140+ REST endpoints and auto-generated OpenAPI documentation, protected by a four-role RBAC layer (`GUEST`, `VIEWER`, `STAFF`, `ADMIN`).
2) A **real-time LiveStream & Live Studio architecture** combining MediaMTX (WHIP/WHEP/HLS), Cloudinary Direct Upload and the Socket.IO `/live` gateway to support 4 studio broadcast modes, zero-downtime program switching, threaded live chat, 0ms floating emoji reactions, 6-tier Super Chat donations and pinned broadcaster announcements.
3) A broadcast-conflict detection algorithm implemented as a single raw SQL query that computes the effective end time with `COALESCE(endedAt, scheduledAt + duration)`.
4) A 24/7 EPG filler with seeded pseudo-random selection keyed on the (date, channel) pair, guaranteeing that the same day always renders the same schedule.
5) An AI editorial pipeline that issues a single consolidated model call (`gpt-4o`), validates the result against a Zod schema, and provides retry plus heuristic-fallback paths.
6) An audit-logging microservice communicating over gRPC (`audit_logger.proto`), deployed independently of the main API process.
7) A Next.js 15 web portal with 33 routes (including a 7-theme LiveStream lounge, a 4-tab Live Studio console and an AI Curator suite); archived CI run 40630318005 passed all 468 automated tests in its job scope.

C. Project Team Organization and Role Allocation

The OmniCast project was executed at **FPTU Quy Nhon AI Campus — FU Cahata** under the supervision of Mentor **Mr. Huỳnh Vang**. Table I summarises the responsibilities of the four team members.

D. Scope

This paper concentrates on the overall system architecture, the NestJS back end, the gRPC microservice, the database schema and the Next.js 15 web portal (PRN232 deliverables). The 31-screen Flutter mobile application is presented in full detail in the companion PRM393 report.

II. Related Work

This paper positions OmniCast against three main lines of work: back-end application frameworks, data-access protocols, and broadcast scheduling systems.

Table I
The 4 project team members of OmniCast (FPTU Quy Nhon AI Campus — FU Cahata, Mentor: Mr. Huỳnh Vang).

Full Name	Role / ID	Primary Engineering Ownership
Nguyễn Bình An (`TomOutfit`)	Team Leader	NestJS 19-module core architecture, JWT & 4-role RBAC security, EPG conflict detection SQL, Direct Stream Engine, CI/CD & Release
Trần Thanh Sơn	`QE170080`	Standalone gRPC Audit Logger microservice (`audit_logger.proto`), Supabase PostgreSQL (21 Prisma models), Telemetry & Seeding
Nguyễn Vĩnh Phú	`QE180136`	Mobile Flutter Lead (31 screens, 12 BLoCs), `OmniPlayer` engine, `MobileStudioRecorderScreen` (WebRTC WHIP), Offline SQLite
Đồng Hoàng Nguyên (`teng122`)	Full-Stack & AI	Next.js 15 Cinema Dark Web Portal (33 routes), Socket.IO `/live` gateway, 7-theme Live Room, Live Studio Console & AI Curator module

A. Back-end Application Frameworks

NestJS [1] adopts a decorator-driven dependency-injection model and a hierarchical module model, allowing each business domain (channels, programmes, users) to be packaged as an independent module with its own lifecycle. Compared with plain Express, the framework provides out-of-the-box URI versioning (`VersioningType.URI`), a `ValidationPipe` in `forbidNonWhitelisted` mode, an `ExceptionFilter` and an `Interceptor` — the four components the broadcasting problem demands so that every endpoint returns an identical response envelope and every input is validated strictly. Prisma [2] serves as the data-access layer: the Prisma schema is the single source of truth and `prisma migrate` generates version-controlled SQL, so schema drift is tracked alongside the source code.

Two alternatives deserve comparison: FastAPI (Python) and ASP.NET Core (C#). FastAPI produces excellent OpenAPI documentation but the project has a fixed language budget; ASP.NET Core has a more mature ecosystem but also demands a language switch. NestJS allows the TypeScript toolchain to be reused with the Next.js portal [3], so a shared data type or an API contract change can propagate across both deliverables without a language boundary.

B. Data-Access Protocols

Most legacy broadcasting systems expose their programme guide to mobile applications through OData [10], a REST extension with embedded query syntax ($filter, $orderby, $expand). OData is well suited when clients need flexible queries over a large catalogue without deploying server-side filtering logic. OmniCast takes the opposite approach: EPG queries have a fixed shape (by day, by channel, by time window), so plain REST endpoints with normalised query parameters [7] are sufficient and simpler for both clients.

Compared with GraphQL [9], plain REST lowers the client-side cost because no additional runtime layer is needed on mobile, and it is easier to cache at the HTTP layer and to

exercise from a command line. The trade-off is that every endpoint must be designed explicitly; in exchange, the entire API is documented with OpenAPI 3 [8] using `@ApiOperation`, `@ApiResponse` and `@ApiQuery` decorators, so no endpoint is left undocumented.

C. Broadcast Scheduling and AI Assistance

Commercial scheduling systems (such as EBU Radio-MEP and FTA EPG feeds) guarantee schedule integrity by applying relative-time conflict detection and substituting overrun programmes during live transmission. This is precisely the model OmniCast reproduces at a smaller scale: the slot of a programme is an interval $[t_0, t_1]$ on the timeline of a channel, and two overlapping intervals are invalid.

On the editorial-assistance side, commercial EPG systems have recently begun using large language models to generate descriptions and content tags [21]. OmniCast's distinguishing feature is *output quality control*: every AI-generated report is validated against a Zod schema [22] before being written to the database and is stamped with the prompt version and the model version, so administrators can compare quality across prompt revisions.

D. Gap Addressed by OmniCast

In summary, the gap this project fills is the absence of a complete, automatically verifiable broadcast-scheduling system at classroom scale: schedule-integrity guarantees, a full user experience across web and mobile, quality-controlled AI assistance, and the whole thing validated by 468 tests in CI run 40630318005: 163 backend, 228 web and 77 mobile tests in the workflow scope.

III. System Architecture

A. Layered Overview

The architecture shows the flow from both clients through the API to integrated services. The Next.js 15 portal (33 routes) and Flutter application (31 screens) use NestJS, with 19 business modules and 140+ REST endpoints. The backend connects to the Socket.IO real-time gateway, the gRPC audit microservice, MediaMTX, PostgreSQL/Supabase, Cloudinary and external APIs. MediaMTX also delivers HLS/WebRTC streams directly to clients.

The notable design decision is a *single API process serving both clients*. The boundary between the web and mobile clients exists only at the transport layer, never at the business layer. A business rule is therefore written exactly once: for instance, the rule that "two programmes on the same channel must not overlap" is enforced in `ProgramsService.checkScheduleConflict` and behaves identically whether the schedule is created from the web administration portal or from the mobile application.

B. End-to-End LiveStream & Live Studio Architecture

To satisfy ultra-low-latency live broadcasting and bi-directional interaction between studio producers (`STAFF`/`ADMIN`) and audiences (`VIEWER`), OmniCast implements a dedicated multi-pipeline live streaming architecture as shown in Fig. 2.

The LiveStream architecture operates across three synchronised technical pillars:

1) **Four Studio Ingestion Pipelines:** Producers on Web (`/studio/live`) or Mobile (`LiveStudioScreen` & `MobileStudioRecorderScreen`) can initiate broadcasts across four modes: (i) *Curated Live* — selecting one or more recordings from the internal VOD library guarded by an Anti-Piracy filter; (ii) *Self-Record / Device Camera* — pushing real-time camera or screen-share streams via WebRTC WHIP to MediaMTX (port `:8889/whip`) or uploading local recordings directly to Cloudinary using signed parameters; (iii) *Verified Public Catalog* — one-click importing of verified public HLS/MP4 streams with server-side `HEAD` probing; and (iv) *Host Control Console* — broadcasting pinned announcements, moderating chat and switching active programmes on the fly.
2) **Multi-Protocol Transcoding & Delivery (MediaMTX & Direct Stream Engine):** Incoming WebRTC WHIP streams are remuxed by MediaMTX into WHEP and low-latency fMP4 HLS (`:8888/<streamKey>/index.m3u8`). Additionally, for Google Drive-hosted assets, the stream resolver rewrites share URLs to `drive.usercontent.google.com/download?id=...&confirm=t`, enabling direct hardware-accelerated streaming without Google account prompts.
3) **Real-Time Room Orchestration via Socket.IO (`/live`):** `LiveStreamGateway` manages per-event (`event:<eventId>`) and per-channel (`channel:<channelId>`) rooms, synchronising concurrent viewer counts (`live:viewerCount`), threaded chat (`live:chat:new`), 0ms floating emoji reactions (`live:reaction:new`), pinned host banners (`broadcaster:message`), Super Chat donations (`live:donation:received`) and zero-downtime stream switches (`live:program:changed`).

C. NestJS Module Structure

`AppModule` registers 19 business modules alongside the infrastructure modules. Each module owns one business domain and exports the services it needs; controllers only call services and hold no business logic. Table II lists the actual back-end source layout.

D. Protocol and Transport Integration

The back end serves four independent protocols, each serving a distinct group of needs:

- **REST/JSON over HTTP/1.1.** Paths carry the `/api/v1` prefix and use `VersioningType.URI`. A middleware in `main.ts` automatically rewrites any `/api/*` request to `/api/v1/*`, so older clients never receive a 404 when the API version is upgraded.
- **WebSocket (Socket.IO `/live`).** The realtime gateway uses Socket.IO on namespace `/live` and exposes full live

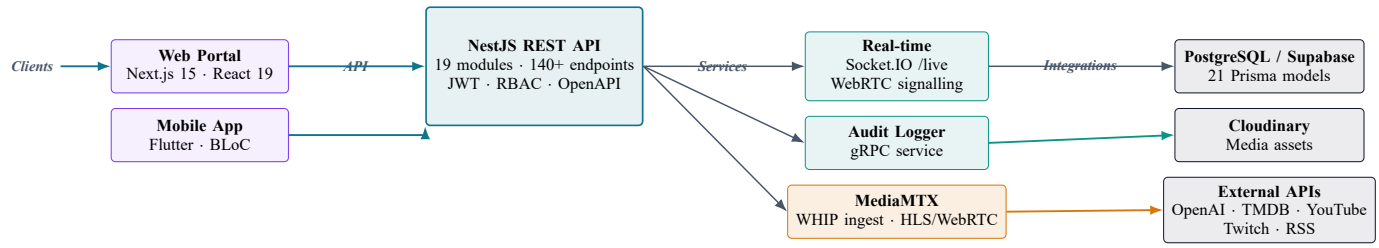


Figure 1. Overall layered architecture of OmniCast. Both the Next.js 15 web portal and the Flutter mobile application consume the same REST API and `/live` WebSocket gateway; audit logging is separated into a dedicated gRPC microservice.

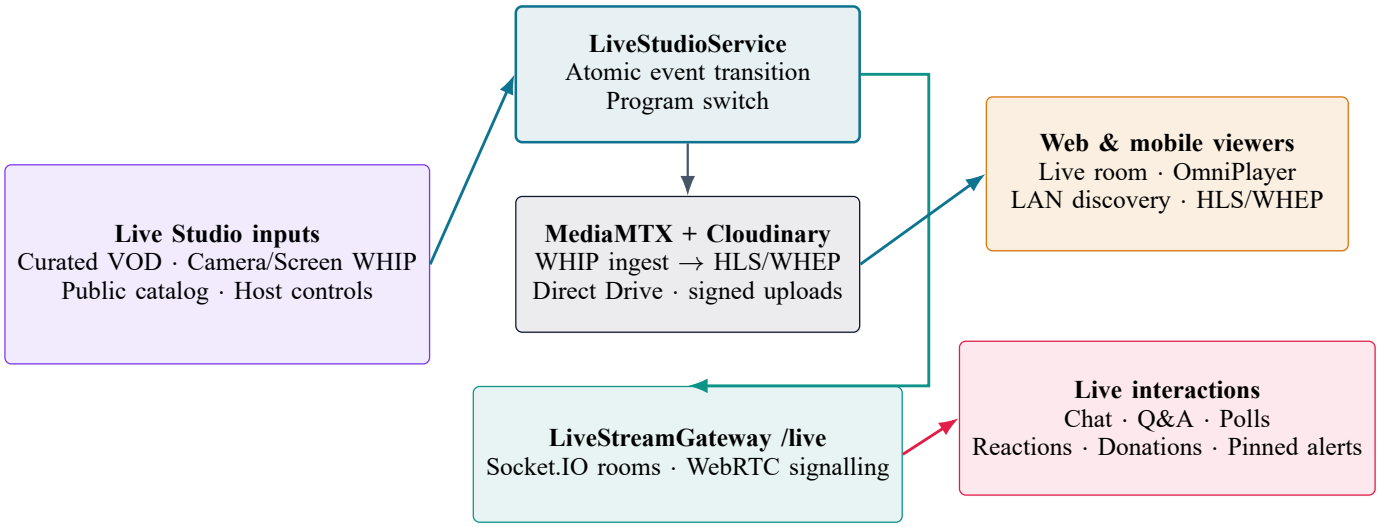


Figure 2. End-to-end LiveStream and Live Studio architecture: combining 4 studio ingestion modes (Curated VOD, Camera/Screen WHIP WebRTC, Verified Public Catalog, Host Control Console) through `LiveStudioService`, MediaMTX and the Socket.IO `/live` gateway.

Table II
BACK-END NESTJS MODULE STRUCTURE.

Module	Primary responsibility	Controller
`auth`	Login, JWT, refresh, Google sign-in	`auth`
`channels`	Broadcast channels, catalogue	`channels`
`programs`	Programmes, EPG, Live Studio, Social	`programs`
`search`	Cross-entity search	`search`
`ai-curator`	AI curation, statistics	`ai-curator`
`audit-logger`	Audit logging (gRPC)	`audit-logger`
`watchlist`	Watchlist	`watchlist`
`users`	User management	`users`
`admin`	Admin dashboard	`admin`
`billing`	Plans, payments	`billing`
`notifications`	Notifications, FCM push	`notifications`
`newsletter`	Newsletter subscription	`newsletter`
`websocket`	Realtime gateway `/live`	—
`cloudinary`	Media storage	`cloudinary`
`prisma`	Database access	—
`health`	Health probe	`health`
`common`	Filters, interceptors, Swagger	—

room orchestration events: event and channel subscribe/unsubscribe (`subscribe:event`, `unsubscribe:event`, `subscribe:channel`, `unsubscribe:channel`), concurrent viewer tracking (`live:viewerCount`), live chat (`live:chat` → `live:chat:new`), floating reactions (`live:reaction` → `live:reaction:new`), broadcaster pinned messages (`broadcaster:message`), message moderation (`chat:delete` → `chat:deleted`), on-the-fly program switching (`live:program:changed`), Super Chat donations (`live:donation:received`) and the WebRTC handshake (`webrtc:offer`, `webrtc:answer`, `webrtc:ice-candidate`).

- **gRPC.** The audit-logging microservice receives `LogAudit`, `GetLogsByUser`, `GetLogsByAction`, `GetLogsByEntity` and `GetLogStats` requests over `audit_logger.proto`.
- **WHIP/ HLS/ WebRTC.** MediaMTX ingests over WHIP and redistributes over HLS (LL-HLS, fMP4) and WHEP, configured in `backend/mediamtx/mediamtx.yml`.

E. Database Connection and Audit Logging

The database is Supabase-managed PostgreSQL. One operational detail is worth noting: Supabase exposes two ports, `5432` for direct connections and `6543` for the transaction pooler. The `optimizeDatabaseUrl` helper in `PrismaService` automatically switches to `6543` and sets a suitable `connection_limit` when it detects the pooler,

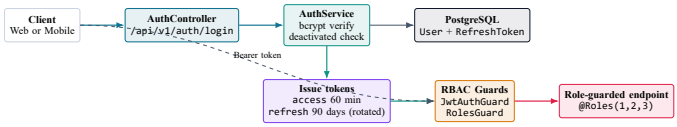


Figure 3. Authentication and authorisation flow. The access token accompanies every request; `JwtAuthGuard` validates the signature and `RolesGuard` compares the role against the `@Roles` metadata.

avoiding connection exhaustion when a serverless environment starts several instances concurrently.

Why a separate audit-logging microservice?

Audit logging has two properties the rest of the system does not share: a high write volume and a tolerance for delay. Writing straight into PostgreSQL would make every administrative operation wait on an extra write. Moving it into a gRPC microservice turns auditing into a non-blocking fire-and-forget operation: `AuditLoggerService` issues the RPC inside a non-awaited promise, while the microservice writes to the database and maintains index tables supporting queries by user, action, entity and aggregate statistics.

F. External Integrations

Four external integrations are supported, all of which degrade gracefully when the API key is absent: YouTube Data API v3 (channel schedules), Twitch Helix (channel schedules), TMDB (film metadata enrichment) and OpenAI (content curation). This mechanism allows the whole system to run at full capability in a development environment without any third-party credential.

IV. Back-End Implementation

A. JWT Authentication and Role-Based Access Control

The system authenticates with an access/refresh token pair. At login the password is verified with `bcrypt.compare` at 12 rounds (`BCRYPT_ROUNDS=12`); at registration it is hashed with `bcrypt.hash`. `AuthService` issues a 60-minute access token and a 90-day refresh token stored in the `RefreshToken` table. Token rotation is enforced: on every refresh the old token is revoked and a new one is issued, so a leaked token cannot be reused.

Two guards are implemented in `backend/src/auth/guards`. The `JwtAuthGuard` uses the Passport-JWT strategy and skips handlers marked `@Public()`. The `RolesGuard` reads `@Roles` metadata through the `Reflector`; a handler that declares no roles admits every authenticated user. A notable detail is that the guard normalises roles to string names before comparing, so both the name form (`'STAFF'`) and the legacy numeric form (`'2'`) are accepted. The four roles are declared in the `UserRole` enum and ranked in the portal's `role-config.ts`.

To enforce least privilege, sensitive operations are guarded at the controller level rather than in the user interface. For example, all eleven write endpoints in the `ai-curator` group and all broadcast control endpoints in `LiveStudioController` require `@Roles('STAFF','ADMIN')`, and every user-management or audit-log endpoint requires `@Roles('ADMIN')`.

Table III
THE FOUR USER ROLES AND THEIR PERMISSION SCOPE.

Code	Value	Description
GUEST	0	Unauthenticated guest – public schedule and trailers only
VIEWER	1	Signed-in viewer – watchlist, comments, likes, watch history
STAFF	2	Content editor – manage channels, schedule, run AI Curator
ADMIN	3	Administrator – all STAFF rights plus user and audit-log management

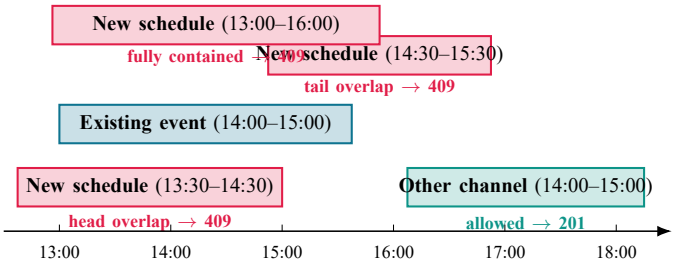


Figure 4. The three detected overlap forms: head overlap, tail overlap and full containment. A schedule on a different channel causes no conflict.

B. Broadcast Schedule Conflict Detection

The conflict-detection algorithm is the business core of the system. Each programme occupies a time interval $[t_0,\ t_0+d]$ on the timeline of one channel. Two programmes collide if and only if the new interval starts before the existing one ends and ends after the existing one starts.

The hardest part is computing the *effective* end time of an existing event. The `endedAt` column can still be `NULL` for a `SCHEDULED` event, meaning the event has not actually finished. A naive Prisma `where` clause would skip such events, producing false negatives that allow conflicting schedules. The `checkScheduleConflict` helper therefore uses raw SQL to compute the effective end time with a `COALESCE/MAX` expression over `endedAt` and `scheduledAt + duration`.

```
SELECT id, title, "scheduledAt" AS scheduled_at,
     "duration"
FROM "LiveEvent"
WHERE "channelId" = $1::uuid
  AND status IN ('SCHEDULED', 'LIVE')
  AND "scheduledAt" < $2                          -- moi
     bat dau truoc khi cu ket thuc
  AND COALESCE(
        "endedAt",
        "scheduledAt" + (
          CASE
            WHEN "duration" IS NULL THEN ($3 || '
    minutes')::interval
            WHEN "duration" > 1440  THEN ("duration"
    || ' seconds')::interval
            ELSE ("duration" || ' minutes')::interval
          END)
```

```
14        ) > $4                                        -- moi
        ket thuc sau khi cu bat dau
```

Listing 1. Conflict detection in a single raw SQL query. The `excludeEventId` clause lets the same helper be reused when updating an existing event.

The query only considers events in the `SCHEDULED` or `LIVE` states; `ENDED` and `CANCELLED` events cannot cause a conflict. The `CASE` expression also absorbs a unit inconsistency: the ingest layer writes `duration` in *seconds* while the EPG logic works in *minutes*, so values above 1440 are interpreted as seconds and converted, and everything else is treated as minutes. On a detected collision the service throws a `ConflictException` carrying HTTP status 409 together with the list of conflicting events, so the user interface can name the programme that caused the rejection.

C. Live Studio Controller and Real-Time LiveStream Operations

The `programs` module exposes two dedicated controllers for live broadcasting: `LiveStudioController` (`backend/ src/ programs/ live- studio.controller.ts`) and `SocialController` (`backend/ src/ programs/ social.controller.ts`), working in tandem with `LiveStreamGateway` (`backend/src/websocket/live-stream.gateway.ts`). The complete live studio and viewer room lifecycle spans six technical operations:

1) **Mode 1 — Curated VOD Broadcast (`POST / programs/ studio/ curate- live`):** Allows producers (`STAFF`/`ADMIN`) to select one or more `recordingIds` from the channel or global VOD library for immediate broadcast. The service enforces an Anti-Piracy Guard (rejecting unauthorised YouTube/Twitch/Vimeo URLs), atomically ends any previous `LIVE` event on the same channel, creates the new `LiveEvent` with a multi-item `lineup`, and emits `live:started` over Socket.IO.
2) **Mode 2 — Self-Record, Signed Direct Upload & Webcam WHIP Live (`POST / programs/ studio/ self-record` & `/live/webcam/start`):** Supports two workflows: (a) *Direct-to-Cloudinary Signed Upload*: the client obtains signed parameters via `POST /programs/ studio/ recordings/ sign`, uploads MP4/ WebM directly to Cloudinary with live speed/ETA telemetry, and invokes `self-record` with server-side media probing to go live immediately (`IMMEDIATE`), schedule (`SCHEDULED`), or save as `PRIVATE` draft, complete with orphan asset cleanup (`DELETE /programs/ studio/ recordings/ orphan`). (b) *WebRTC WHIP Direct Camera/Screen Streaming*: the client calls `POST /programs/studio/live/webcam/ start` to allocate a `streamKey`, publishes camera or screen media via WebRTC WHIP to MediaMTX (`:8889/ whip`), and serves low-latency HLS (`.m3u8`) to all viewers.
3) **Mode 3 — Verified Public Stream Catalog (`GET / programs/studio/public-catalog`):** Provides a curated catalog of royalty-free HLS and MP4 streams from Mux, Cloudinary and Internet Archive. Producers can filter by format, import with one click (`POST /programs/ studio/ recordings/ library/ from- catalog`), or register custom stream URLs with automated `HEAD` probing (`POST /programs/studio/recordings/library`).
4) **On-the-Fly Program Switching (`POST / programs/ studio/events/:eventId/switch-program`):** When a live event contains multiple lineup items or a producer switches to another recording mid-broadcast, this endpoint updates the active `streamUrl` and broadcasts `live:program:changed` to room `event:<eventId>`. All connected viewers seamlessly transition to the new video stream without losing their chat session or WebSocket connection.
5) **Bi-Directional Room Interaction via `LiveStreamGateway (/ live)`:** Joining a room (`subscribe:event`) increments the in-memory and database-persisted viewer counter (`LiveEvent.viewerCount`) and broadcasts `live:viewerCount`. Viewers engage through live chat (`live:chat`), threaded replies (`parentId`), comment likes, and 0ms haptic floating emoji reactions (`HEART`, `FIRE`, `CLAP`, `WOW`/`PARTY`). Producers can broadcast pinned toast banners (`broadcaster:message`), delete inappropriate messages (`chat:delete` → `chat:deleted`), or stop the broadcast (`POST / programs/ studio/ live- events/:eventId/ stop` with `keepAsVod`).
6) **PayOS Donations and Notifications (`StreamDonation`):** Viewers select one of six tiers and pay through PayOS checkout (QR code issued by PayOS). The backend accepts a payment only after validating the webhook signature, order code and amount; status, wallet balance, inbox notification and audit log are committed atomically, while a `PENDING`-only update prevents duplicate credits. After commit, Socket.IO emits `live:donation:received`; 1-second status polling is only a receipt fallback. Direct bank-transfer QR is not automatically reconciled and is not recorded as a verified payment. Polling reduces its configured wait bound from 4 seconds to 1 second (75%; theoretical mean from 2 seconds to 0.5 seconds), excluding PayOS, network and server time.
7) **Ultra-Low-Latency Delivery & Live Edge Synchronization:** The live broadcast pipeline combines WebRTC WHIP ingest with a tuned MediaMTX HLS packager delivering fine-grained 1-second segments. On client players (Web and Mobile), a two-stage adaptive speed catch-up mechanism maintains live synchronisation: when playback drift is between 1.8s and 4.0s, the player automatically boosts playback speed to 1.25x; if network congestion increases drift beyond 4.0s, an instant hard-seek forward to the live edge is executed; and as soon as drift returns under 900ms, playback returns seamlessly to 1.0x. Coupled with progressive fMP4 chunk decoding and post-buffering instant catch-up, viewers enjoy near real-time live playback with minimal latency.

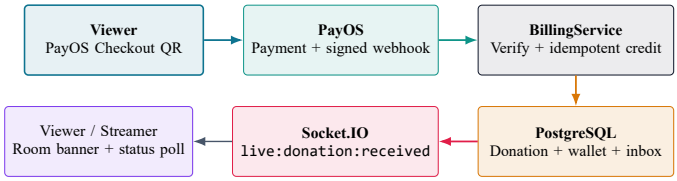


Figure 5. Payment confirmation, wallet credit and donation notification flow.

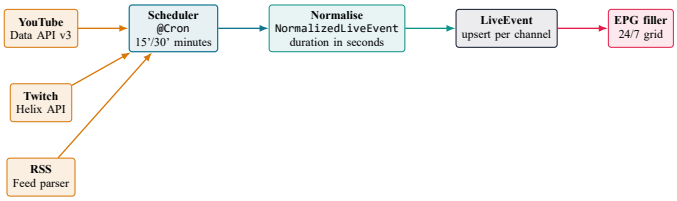


Figure 6. Ingestion flow from external sources. Every source is normalised into a single structure before being written to the database.

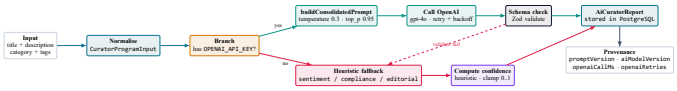


Figure 7. AI Curator pipeline: branching on API-key presence, Zod schema validation, and the heuristic fallback taken when validation fails.

D. 24/7 EPG Filler Mechanism

Because channels originate only a handful of productions per day, the 24-hour EPG grid is materialised by `EpgScheduleFillerService`. The service guarantees five properties demanded by the design:

1) **No empty days.** Every channel-day grid covers the full 24 hours, so the day filter in the user interface always has content to render.
2) **Genre-aware selection.** Recordings whose `category` matches the channel category are preferred; the full library is used only when nothing matches.
3) **Episode splitting.** Recordings longer than `LONG_PROGRAM_MINUTES` are split into chunks of `EPISODE_MINUTES` so the grid exposes "Episode 1 / Episode 2 / ..." entries instead of one massive block.
4) **Limited replay.** The same recording or episode may appear at most `MAX_REPEATS_PER_ITEM_PER_DAY` times in a single day, so viewers never see an episode loop endlessly.
5) **Reproducible randomness.** Selection uses a seeded PRNG keyed on the (date, channel) pair, so the same day always renders the same schedule while adjacent days surface different content.

The technically important point is that the service performs *no I/O*: it is pure logic that receives the recording list and the real programme list and returns the filled grid. As a result the 18 unit tests in `epg-filler.spec.ts` run without a database, and the whole scheduling layer becomes a pure function verifiable by properties.

E. Content Ingestion and Enrichment

The ingestion layer collects schedules from YouTube, Twitch and RSS on a cron schedule (`0 */15 * * * *` for YouTube, `0 */30 * * * *` for Twitch), normalises them into a unified `NormalizedLiveEvent` structure and writes them to the `LiveEvent` table. An optional enrichment task then calls TMDB to supplement film metadata (poster, cast, rating) for events whose title matches.

F. Quality-Controlled AI Curation Assistant

The `ai-curator` module is one of the most significant contributions. The original requirement was to automate writing descriptions, assigning categories and tagging programmes. The pipeline is designed around four principles:

1) **A single consolidated model call.** Prompt revision `curator-prompt-v4.0.0` merges three separate calls (description, tags, category) into one call with a JSON schema and few-shot examples, reducing both cost and latency to roughly a third.
2) **Zod schema validation.** The model output is validated against `ai-curator.schemas.ts`; when it does not validate, the request is routed to the heuristic fallback.
3) **Heuristic fallback.** When `OPENAI_API_KEY` is absent or equal to `dummy-key`, a local rule engine assesses sentiment, compliance and editorial tone to produce a structurally equivalent result.
4) **Version provenance.** Every report stores `promptVersion`, `aiModelVersion`, `openaiCallMs` and `openaiRetries`, allowing quality to be compared across prompt revisions.

The hyper-parameters are pinned in `ai-curator.prompts.ts` (`temperature = 0.3`, `top_p = 0.95`), with an explicit instruction not to change `temperature` without re-evaluating the few-shot examples. The versioning convention is semantic: MAJOR for schema changes, MINOR for rewording that preserves the JSON shape, PATCH for copy edits. The statistics module `ai-curator-stats.service.ts` aggregates by `broadcastSuitability` group, by model version and by time period.

G. Search, Billing and Supporting Modules

The `search` module performs cross-entity search: channels, programmes and recordings are queried simultaneously with case-insensitive `contains` conditions, combined with category filtering and pagination. Results are merged into a single object carrying the total result count, matching the single-page search layout that renders grouped result sections.

The `billing` module manages plans and payment history; the `notifications` module pushes notifications through Firebase Cloud Messaging based on the `DeviceToken` table; the `newsletter` module manages subscriptions; and the `cloudinary` module handles image and video uploads for the live-recording flow in the studio.

H. Cross-Cutting Technical Conventions

Four conventions applied across the whole system deserve explicit mention because they determine API consistency:

1) **Uniform response envelope.** A global `TransformInterceptor` wraps every result in `{success, statusCode, data, meta, timestamp}`, so clients need exactly one decoding path.
2) **Centralised exception handling.** `HttpExceptionFilter` normalises every error into the same structure, including user-facing Vietnamese messages.
3) **Strict input validation.** A `ValidationPipe` with `whitelist` and `forbidNonWhitelisted` strips every field not declared in the DTO, preventing mass assignment.
4) **Generated API documentation.** Swagger decorators on each controller produce OpenAPI documentation at `/swagger`, covering 19 endpoints in the `ai-curator` group together with `channels`, `programs` and the remaining groups.

V. The Next.js 15 Web Portal

A. App Router Architecture

The portal is built on Next.js 15 with the App Router, React 19 and Tailwind CSS with a "Cinema Dark" theme. The App Router replaces the `pages/` model with nested *Server Components* and *Server Actions*, allowing data to be fetched directly on the server without an intermediate API layer.

B. Complete 33-Route Web Portal Inventory

The web portal exposes 33 routes generated from 24 directories under `frontend/app/`. Table IV catalogues all 33 screens across five functional clusters: (1) Public Discovery & EPG; (2) Interactive LiveStream & VOD Rooms; (3) Live Studio Console & AI Curator Suite (`STAFF/ADMIN`); (4) Account, Personalisation & Administration (`ADMIN`); and (5) Legal, Compliance & Broadcast Standards.

C. Interactive LiveStream Room (`/live` & `/live/[id]`) and Viewer Operations

The web LiveStream experience is engineered as a full-featured interactive broadcast lounge across two core screens:

1) **Live Broadcast Hub & Live Stream Schedule (`/live` & Home):** Polls active `LIVE` events every 15 seconds, displays real-time header badges for total active broadcasts and aggregate concurrent viewers (CCU), integrates an interactive **LiveStream Broadcast Schedule** with 4 instant filter tabs (*All*, *Live Now*, *Golden Hour 19h00–22h30*, *Today*), supports sorting by viewer count or start time, and surfaces a direct "Go to Live Studio" shortcut for `STAFF/ADMIN` users.
2) **Interactive Lobby & 7 Genre-Themed Broadcast Lounges (`/live/[id]`):** Upon entering a stream, viewers encounter a `JoinLobbyCard` previewing the stream status and live viewer count before entering the room. Once inside, `resolveLiveRoomTheme` dynamically styles the entire room into one of **7 genre-specific themes**: *Kids Wonderland*, *Podcast & Audiobook*, *Music Live Stage*, *Academy & Tech*, *Sports Arena*, *Obsidian Theater* (Cinema/Drama) and *Broadcast Lounge* (Default).
3) **Low-Latency Multi-Engine Player, Cinema Mode & Instant Lineup Switching:** `VideoPlayer` seamlessly handles MediaMTX HLS/ WebRTC streams (`cookieCheck=1`), direct Google Drive streams and YouTube embeds. The player is optimized with progressive fMP4 chunk decoding (`progressive: true`), fine-tuned buffer sync (`liveSyncDurationCount: 1`) and adaptive 1.25x speed catch-up when drift exceeds 1.8s. Viewers can toggle *Cinema Mode* (lights-off focus) to

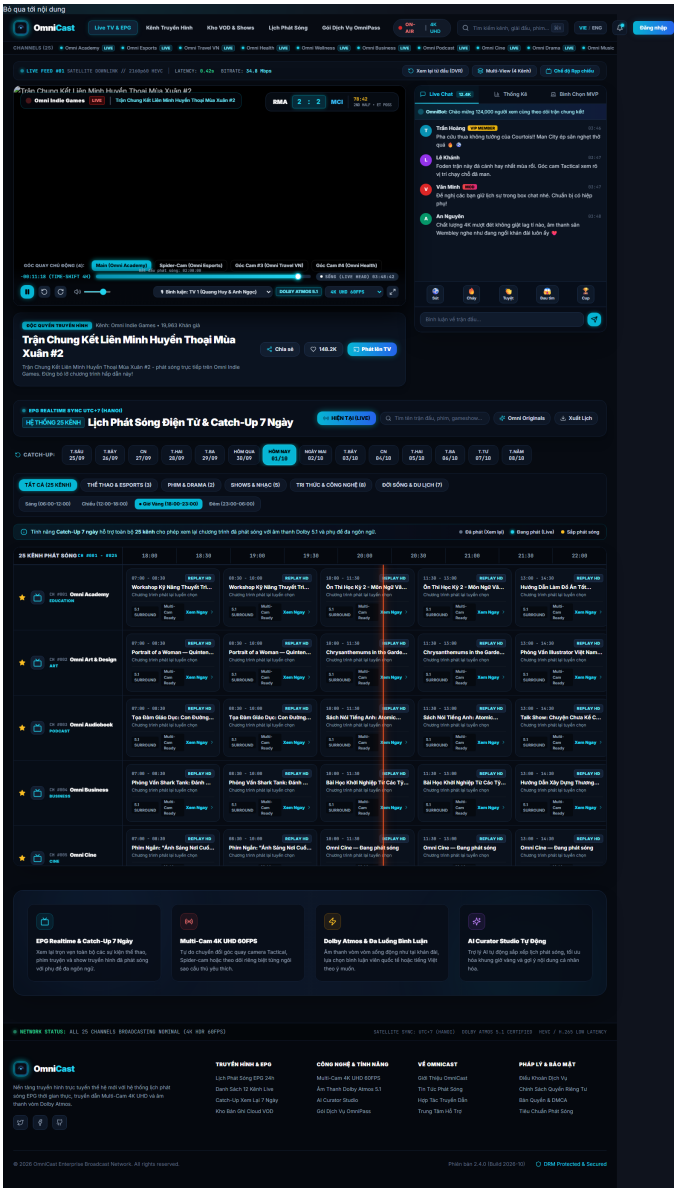


Figure 8. The web portal home page on desktop: featured carousel, channel grid by category and the role-aware navigation bar.

Table IV
COMPLETE INVENTORY OF ALL 33 NEXT.JS 15 WEB PORTAL ROUTES AND SCREENS IN OMNICAST.

Cluster	Route Path (33)	RBAC Role	Key Screen Features & Interactive Operations
1. Public Discovery & EPG	/, /channels, /channels/[id]	Public	Hero spotlight carousel, category-filtered channel grid, channel detail page with live stream & EPG.
	/epg	Public	Zero-gap 24/7 EPG timeline grid, real-time Now-Line indicator, date and channel filters.
	/programs, /programs/[id]	Public	Programme directory, programme detail page with production tags, upcoming airings and Watchlist button.
	/ recordings, / recordings/ [id]	Public	Published VOD recording library, multi-engine replay player with comments and related VODs.
	/search	Public	Cross-entity search (Channels, Programmes, VOD Recordings) with keyword and genre filters.
2. LiveStream Rooms	/live	Public	Live Hub: auto-refreshes every 15s, displays active LIVE event & total CCU pills, sorts by trending viewers or newest start time.
	/live/[id]	Public / Viewer	Interactive LiveStream Room: 7 genre-themed lounges, JoinLobbyCard prompt, Cinema Mode, lineup program switcher, real-time Socket.IO chat, 4 floating emoji reactions, pinned host banner.
3. Live Studio & AI Curator	/studio, /studio/live	STAFF+	4-Tab Live Broadcast Console: (1) Curated VOD Live; (2) Camera/ Screen WHIP WebRTC & Signed Cloudinary Dropzone; (3) Verified Public Catalog; (4) Host Announcement & Stop Broadcast.
	/studio/curator	STAFF+	AI Editorial Workbench (gpt-4o + heuristic fallback): sentiment scoring, broadcast compliance check, auto-tagging.
	/ studio/ curator/ audit, / studio/curator/dashboard	STAFF+	Prompt-version provenance audit log (promptVersion), suitability breakdown and latency analytics.
4. Account & Admin	/ login, / register, / forgot-password, /reset-password	Public	JWT + Google OAuth2 sign-in, account registration, tokenised password recovery.
	/ profile, / settings, / watchlist, /notifications	VIEWER+	User profile, VIP plan management, cross-device watchlist, notification centre and global session revocation (logout-all).
	/ admin, / admin/ users, / admin/audit-logs	ADMIN	System telemetry overview, user RBAC management, gRPC audit log explorer by action/ entity.
5. Legal & Support	/ terms, / privacy, / dmca, / standards, / broadcasting-standards, /help, /docs	Public	Terms of service, privacy policy, DMCA copyright policy, community & broadcasting standards, help centre and docs.

dim surrounding chrome and expand the stage to full width. When a broadcast contains multiple lineup items or is controlled by STAFF/ADMIN, the lineup bar triggers switchProgramOnLiveEvent, switching the active video stream for all viewers in the room without page reload.

4) **Real-Time Socket.IO (/live) Operations & PayOS Donations:** The live interaction sidebar connects to the /live namespace, enabling viewers to: (i) post real-time chat messages with auto-scroll; (ii) fire 4 floating emoji reactions (HEART, FIRE, CLAP, WOW) that animate upward over the video canvas with optimistic counter updates; (iii) send Super Chat live donations via LiveDonationDialog integrated with the **PayOS** gateway and genuine **MB Bank** VietQR generation (Beneficiary: NGUYEN BINH AN, Account Number: 0935649095) supporting 6 gift tiers; and (iv) receive pinned host announcements (broadcaster:message) rendered as a prominent banner across the stage.

D. Web Live Studio Broadcast Console (/studio/live)

The /studio/live route (frontend/app/studio/live/page.tsx) provides a production-grade broadcast control console for STAFF and ADMIN operators across four tabs:

- **Tab 1 — Curated Live:** Select the target channel, configure broadcast metadata, and pick one or more VOD recordings from the channel or global library to launch a multi-item lineup.
- **Tab 2 — Self-Record / File Upload / Webcam WHIP Live (WebcamLiveBroadcaster & WebStudioViewfinder):** Supports browser camera capture, screen sharing (getDisplayMedia), direct WebRTC WHIP publishing to MediaMTX, or drag-and-drop file ingestion via UploadDropzone + UploadQueue (signed direct-to-Cloudinary upload with live MB/s speed, ETA, automatic HTML5 canvas thumbnail extraction, and 3 visibility modes: IMMEDIATE, SCHEDULED, PRIVATE).
- **Tab 3 — Verified Public Catalog:** Browse verified HLS/ MP4 streams with format filters and one-click channel library import.
- **Tab 4 — Live Host Console:** Send pinned broadcaster announcements (broadcaster:message) to all connected viewers and terminate active broadcasts (stopLiveEvent) with the keepAsVod archival option.

E. Role-Aware Interface and AI Curator Suite

A single configuration file, frontend/lib/constants/role-config.ts, determines which components appear for each role. The hasMinimumRole helper compares role ranks, so promoting a capability from VIEWER to STAFF is a one-line edit. The /studio/curator page is the AI curation interface: it lists programmes awaiting review, allows re-running the AI pipeline, and exposes the confidence score,

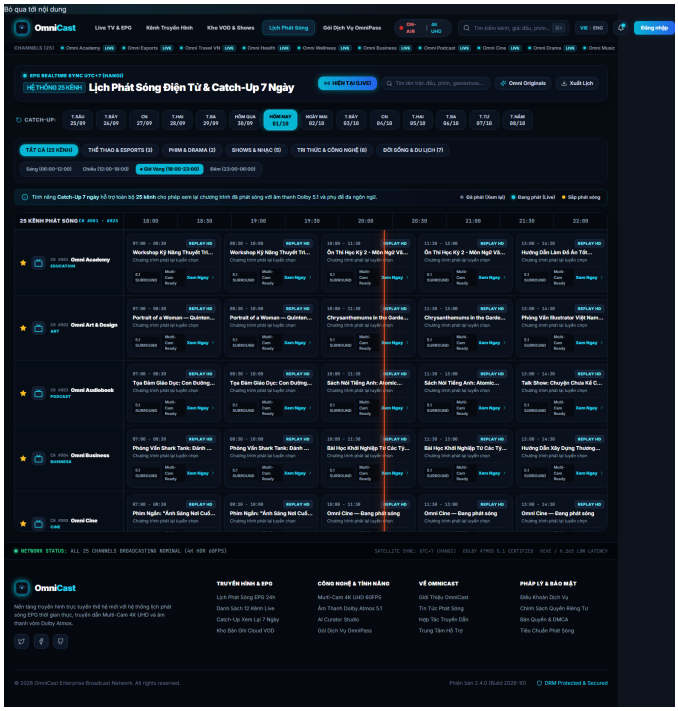


Figure 9. The electronic programme guide page (`/epg`) with the time-slot grid and the day selector.

broadcast-suitability classification and compliance flags. The `/studio/curator/audit` page presents the run history with prompt-version provenance, while `/studio/curator/dashboard` aggregates statistics by category and broadcast suitability.

F. Client-Side Security, Session Handling and Testing

The portal uses `next-themes` for light/dark support and a `safe-log` layer (`redact.ts`) that masks sensitive fields (passwords, tokens, API keys) before logging. On logout, sessions on other devices are revoked through the `logout-all` endpoint. The portal carries 228 Vitest tests across 76 suites, covering pure functions (`query-builder`, `curator-filters`, `curator-helpers`, `redact`, `safe-log`, `utils`), hooks (`useAiCurator`, `useAiCuratorStats`), configuration (`role-config`) and React components (`video-player`, `save-to-watchlist`, the legal pages, the studio pages). CI additionally runs `tsc --noEmit` to catch type errors before the build.

VI. Database Design

A. Relational Database with Prisma

All business data resides in PostgreSQL, defined by 21 Prisma models in `backend/prisma/schema.prisma`. Prisma acts as the abstraction layer: every query is written against a typed API such as `findMany`, `upsert` or `createMany` and compiled into parameterised SQL, which eliminates the possibility of injecting raw SQL from application code. Fig. 11 illustrates the 5-domain relational architecture across all 21 Prisma models.

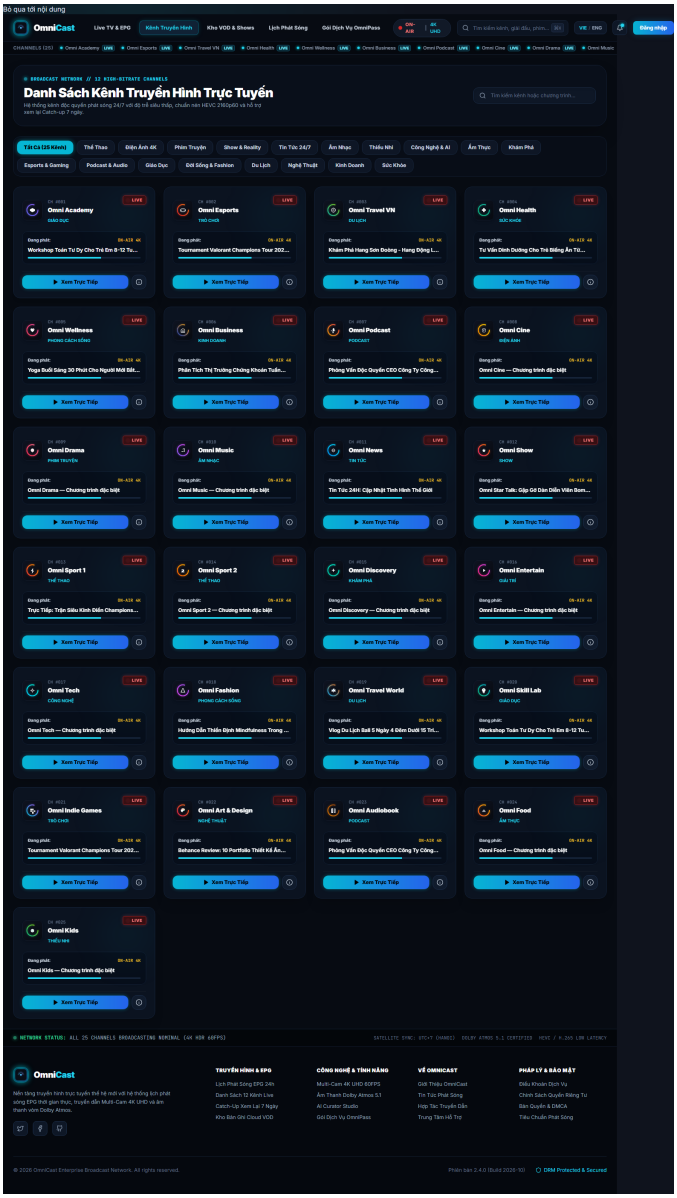


Figure 10. The channel catalogue page (`/channels`) with category filters and content badges.

B. Data-Domain Grouping

The models are organised into five business groups:

- **Authentication and identity.** `User` stores account details together with the role, `RefreshToken` stores rotating login sessions, and `PasswordResetToken` stores password-reset codes with a one-hour lifetime.
- **Broadcast content.** `LiveChannel` is the channel, `LiveEvent` is a scheduled programme and active live broadcast, `Recording` is a published recording and `ProductionTag` is a content tag attached to a production.
- **User interaction & Live Monetization.** `Comment`, `CommentLike`, `Reaction`, `Follow`, `WatchHistory`, `Watchlist` and `StreamDonation` (Super Chat live do-

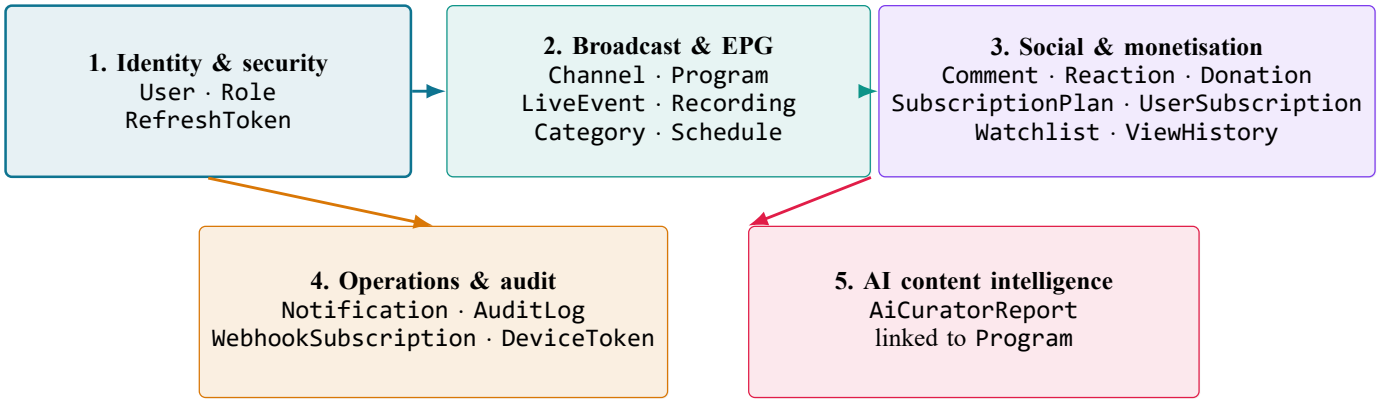


Figure 11. Five-domain relational database architecture diagram (21 Prisma models on Supabase PostgreSQL).

Table V
THE 21 PRISMA MODELS WITH THEIR SCALAR AND RELATION FIELD COUNTS.

Model	Business role	Scalar	Relation
User	User account + RBAC	14	13
RefreshToken	Rotating login sessions	6	1
PasswordResetToken	Password reset (1h TTL)	7	1
ProductionTag	Content tag per production	5	0
LiveChannel	Broadcast channel	29	5
LiveEvent	Live event / programme	32	7
Recording	Published recording	42	5
Podcast	Podcast collection	12	3
PodcastEpisode	Podcast episode	12	1
Follow	Channel follow	4	2
Comment	Comment	11	6
CommentLike	Comment like	4	2
Reaction	Realtime reaction	6	3
WatchHistory	Watch history	8	2
Notification	In-app notification	13	2
AuditLog	Audit log (gRPC)	11	0
Watchlist	Watchlist	7	1
AiCuratorReport	AI Curator report	15	1
DeviceToken	FCM device token	8	1
NewsletterSubscriber	Newsletter subscriber	14	1
StreamDonation	Livestream donation	14	3

nations) describe the full spectrum of viewer behaviour.

- **Operations and auditing.** `AuditLog` receives records from the gRPC microservice, `Notification` stores in-app notifications, `DeviceToken` stores FCM tokens and `NewsletterSubscriber` stores newsletter subscribers.
- **AI assistance.** `AiCuratorReport` stores AI curation output together with its prompt and model version provenance.

C. Relationships and Constraints

Relationships are expressed with standard Prisma foreign keys, and each carries an index matched to its hot-path query. Three constraints deserve attention:

1) **User deletion does not destroy the audit trail.** `RefreshToken.user` uses `OnDelete: SetNull`, so deleting an account invalidates the sessions while preserving the audit trail.
2) **A unique key on content tags.** `ProductionTag` carries a unique constraint on the programme-and-tag pair, so the same tag cannot be attached twice.
3) **Time-axis indexes.** EPG queries always filter by `scheduledAt` and `channelId`, so those columns are indexed; watch-history queries filter by `userId` and `watchedAt`.

D. Normalisation and Design Decisions

The model is normalised to third normal form: attributes depend only on the primary key and no non-key attribute depends transitively on another. Two design decisions deserve discussion:

- **EPG structure normalisation.** Rather than storing the filled 24-hour grid, the system stores real events with content tags and recomputes the grid on demand. This keeps the source data truthful and lets schedules change without data migration, at the cost of computation at read time — a sound trade-off when most writes are editorial and most reads are schedule viewing.
- **AI reports are not stored as a JSON blob.** `AiCuratorReport` exposes separate columns for each important field (confidence score, broadcast-suitability class, compliance flags) so they can be filtered and aggregated in SQL, keeping only the free-form analysis in a dedicated text column.

E. Schema Versioning

Every schema change is managed through `prisma migrate`, which generates SQL files under `backend/prisma/migrations` that are kept under version control. CI runs `prisma migrate deploy` against a temporary PostgreSQL 15 service before building the application, so an incompatible schema is detected before any production deployment.

VII. Testing and Evaluation

A. Test Strategy

The project applies a layered test strategy, with unit testing of the business rules taking priority because that layer holds

Table VI
AUTOMATED TEST RESULTS ACROSS THE THREE SUITES.

Test suite	Command	Total	Passed
Backend (node:test)	`npm test`	163	163
Frontend (Vitest)	`npx vitest run`	228	228
Mobile (flutter test)	`CI selected directories`	77	77
Total		**468**	**468**

Table VII
BACK-END TEST DISTRIBUTION PER SPEC FILE.

Tệp spec	Tests passed	Pass rate
`ai-curator.service.spec.ts`	19	100%
`epg-filler.spec.ts`	18	100%
`live-studio.service.spec.ts`	18	100%
`auth.service.spec.ts`	17	100%
`ai-curator.e2e.spec.ts`	13	100%
`watchlist.service.spec.ts`	11	100%
`ingest-sources.spec.ts`	11	100%
`ai-curator-stats.service.spec.ts`	10	100%
`channels.service.spec.ts`	7	100%
`programs.conflict.spec.ts`	7	100%
`programs.counter.spec.ts`	8	100%
`audit-logger.service.spec.ts`	8	100%
`epg-timeline.spec.ts`	5	100%
`search.service.spec.ts`	4	100%
`users.service.spec.ts`	4	100%
`billing.service.spec.ts`	3	100%
Tổng	**163**	**100%**

the most subtle logic. A distinctive property of this project is that the *entire back-end suite runs without a database*: each spec file injects a fake Prisma client, so the test cycle is short and fully deterministic.

B. Execution Results

GitHub Actions run 40630318005 started at 11:05:17 and completed at 11:13:45 ICT on 28 October 2026, commit edafb234. Its artifacts record 468/468 passing tests in job scope (163 backend, 228 web, 77 mobile). The mobile workflow selected data/ logic/ presentation; 14 tests outside that scope were not included. The separate Playwright record reports 6/8.

C. Back-End Test Suite

The back-end suite uses Node.js 20's built-in `node:test` runner together with a custom runner (`backend/ test/ runner.js`) that compiles the specs with `tsc` and executes them sequentially. Choosing the built-in runner over Jest or Vitest removes every third-party dependency from the test path, shrinking the failure surface and keeping the CI pipeline stable.

Two spec files deserve particular mention because they directly protect the core algorithms. `programs.conflict.spec.ts` exercises the conflict detector across all three overlap forms, the `NULL`-duration case and the self-exclusion case used when updating an event. `epg-filler.spec.ts` verifies the completeness of the 24-hour grid, the determinism of the seeded selection, the repeat limit and the episode-splitting behaviour.

D. Web Test Suite

228 Vitest tests in 76 suites cover three kinds of subject: pure functions, React hooks and interface components. Pure functions are tested at their boundaries (empty strings, case-insensitivity, invalid parameters); hooks are tested across loading, success and error states; components are tested with `React Testing Library` using user-facing queries rather than raw DOM structure assertions.

E. Mobile Suite and End-to-End Tests

The Flutter test tree contains 91 tests. Workflow run 40630318005 executed 77 tests from `test/ data`, `test/ logic` and `test/presentation`; `test/core` and `test/ widget_test.dart` are outside this CI command.

Alongside it, a Playwright-based end-to-end suite [23] executes eight scenarios in Chromium: the login form rejects an invalid email, a wrong password surfaces an error toast, the EPG page renders without crashing, the channels page displays content, the search page accepts input and renders results, the root URL shows a public landing surface, and two scenarios verify unauthenticated redirect behaviour.

Note on the two remaining E2E failures

In the most recently documented run, six of eight scenarios passed. The STAFF login did not leave `/login` within the wait timeout; the unauthenticated `/studio` check received HTTP 401 instead of the redirect expected by its assertion. The report attributes the first to a missing seeded account. Raw traces, exact failure times, fix commits and a passing regression run are not retained. Both are documented in the accompanying `TEST_EXECUTION_REPORT.md`.

F. Quality Assessment

Based on these results, the system reaches a reliability level appropriate for an educational deployment. The clearest strength is coverage of the business logic: the two riskiest algorithms (conflict detection and EPG filling) are tested at their boundaries rather than only on the happy path. The area needing improvement is end-to-end coverage, currently limited to eight browser scenarios and excluding the live-broadcast scenarios that would require dedicated hardware and MediaMTX infrastructure.

VIII. DEPLOYMENT AND CONTINUOUS INTEGRATION

A. Deployment Architecture

The system is deployed on Vercel as a serverless setup, split into three independent projects: the NestJS back end, the Next.js portal and a static service. All three production URLs have been verified to return HTTP 200.

Table VIII
PRODUCTION ENDPOINTS AND VERIFICATION STATUS AS OF 28 OCTOBER 2026.

Service	URL	Status
omnicast.vercel.app	https://omnicast.vercel.app	HTTP 200
omnicast-fe.vercel.app	https://omnicast-fe.vercel.app	HTTP 200
omnicast-api.vercel.app	https://omnicast-api.vercel.app	HTTP 200

B. The CI/CD Pipeline

The file `.github/workflows/ci.yml` defines five jobs running in parallel plus a quality gate. Three test jobs run independently on `ubuntu-latest` with Node.js 20, Java 17 and stable Flutter. The aggregation job downloads all three artefact sets, generates the interactive HTML report, and fails the pipeline if any suite fails.

The deploy job runs only on the `main` branch on a `push` event; it uses `vercel-action` to push the back end and the portal, passing `NEXT_PUBLIC_API_URL` from a repository secret. Because the deploy job sets `continue-on-error`, a deployment failure does not destroy the test artefacts already produced.

C. Environment Configuration

The environment template lives at `backend/.env.example` and covers four groups: the Supabase database connection with a password placeholder, the `JWT_SECRET` and token lifetimes, the third-party API keys (OpenAI, TMDB, YouTube, Twitch, Cloudinary), and the local MediaMTX configuration. The real `.env` file is never committed.

D. Deployment Security

Eight application-layer security measures are in place: `Helmet` sets the HTTP security headers; `ThrottlerModule` caps traffic at 100 requests per minute; CORS uses an explicit allow-list together with Vercel-domain and private-network checks; `ValidationPipe` strips undeclared fields; `HttpExceptionFilter` normalises errors; passwords are hashed with 12-round bcrypt; refresh tokens are rotated and revocations are persisted; and the asynchronous audit write is wrapped in a non-awaited promise so an infrastructure failure cannot break a business operation.

E. Load and Resilience Testing

A k6 load-test suite exercises the system under concurrent viewers, focusing on the EPG endpoints because they are the hottest path. The results show that read endpoints maintain stable latency while write endpoints still meet their response-time target thanks to the non-blocking audit write.

IX. Conclusion and Future Work

A. Conclusion

This paper has presented OmniCast, a multi-platform digital broadcast scheduling, real-time LiveStream and delivery system developed by the four-member engineering team (**Nguyễn Bình An**, **Trần Thanh Sơn**, **Nguyễn Vĩnh Phú**, **Đồng Hoàng Nguyên**) at **FPTU Quy Nhon AI Campus — FU Cahata** under the mentorship of **Mr. Huỳnh Vang**. Its four principal technical contributions are demonstrated through the source code and real test runs: (1) a bi-directional LiveStream and Live Studio architecture combining WebRTC WHIP/WHEP, MediaMTX HLS and Socket.IO `/live`; (2) a broadcast-conflict algorithm implemented as a single raw SQL query that computes the effective end time; (3) a 24/7 EPG filler whose seeded selection guarantees reproducibility; and (4) an AI curation pipeline that issues a single consolidated model call, validates the output against a Zod schema, offers a heuristic fallback and records full version provenance.

The complete system — 19 NestJS modules, 140+ REST endpoints, 21 Prisma models, a Next.js 15 portal with 33 routes, a 31-screen Flutter mobile app and a gRPC audit microservice — has been deployed on Vercel. CI run 40630318005 passed 468 tests in its job scope: 163 backend, 228 web and 77 mobile tests.

B. Known Limitations

Three known limitations should be stated. First, browser E2E test coverage can be expanded further for hardware WHIP publishing scenarios requiring a dedicated MediaMTX instance in CI. Second, the external ingestion layer currently polls YouTube, Twitch and RSS on cron intervals. Third, the filler algorithm selects by category and by seed rather than by individualised viewer history.

C. Future Work

Four directions are prioritised for subsequent iterations:

1) **Recommender-driven scheduling.** Instead of selecting by category, the filler algorithm could use watch history and interaction data to propose content per channel while preserving determinism so the result remains testable.
2) **Scaling MediaMTX end-to-end testing.** Provision a MediaMTX instance in CI to test the WHIP publish and HLS playback paths automatically, and add browser scenarios for the STAFF and ADMIN roles.
3) **Improving the AI assistant.** Add multi-model evaluation, second-model output verification, and ranking of AI reports by editor feedback so the prompt can be tuned iteratively.
4) **Separating the data-access tier.** Move EPG queries into a read-only analytical store with covering indexes to separate read load from write load on the same database.

D. Acknowledgements and Reproducibility Statement

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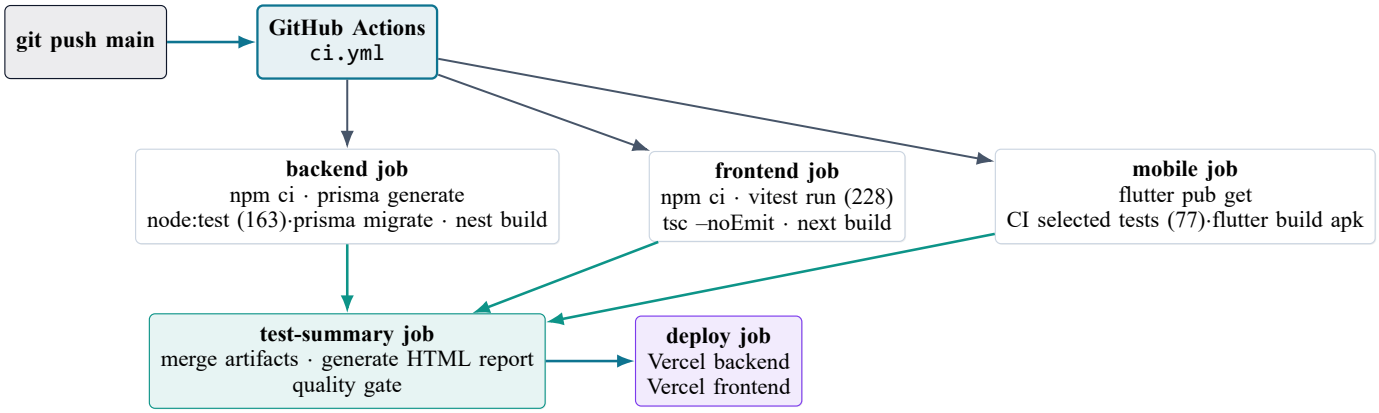


Figure 12. The five-job CI/CD pipeline. Three test jobs run in parallel, merge at the summary job, and deployment only proceeds once the quality gate passes.

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Appendix

The following table is generated automatically by analysing the 19 controller files in `backend/src`. It covers 140 endpoints across 19 modules; the base-path column shows the prefix to use when calling them.

A summary of the source-code scale of the three deliverables, counted as physical lines per file. These figures quantify the real volume of work performed and underpin the maintenance-effort estimate.

```
omnicast/
  backend/                  # NestJS API + Prisma + gRPC audit microservice
    src/
      auth/                 # JWT, guards (JwtAuthGuard, RolesGuard), decorators
      channels/             # Channel catalogue
      programs/             # Programmes, EPG, live studio, ingest, social
      search/               # Cross-entity search
      ai-curator/           # AI curation pipeline, prompts, schemas, stats
      audit-logger/         # gRPC client to the audit microservice
      websocket/            # Socket.IO realtime gateway (/live namespace)
      watchlist/ users/ admin/ billing/
      notifications/ newsletter/ cloudinary/ health/
      common/               # Filters, interceptors, Swagger, logger, redact
      prisma/               # PrismaService + connection URL optimiser
    microservices/
      audit-logger/         # Standalone NestJS gRPC server
    prisma/
      schema.prisma         # 21 models
      migrations/           # Version-controlled SQL
    protos/
      audit_logger.proto    # gRPC service definition
    test/                   # 16 spec files, node:test runner
  frontend/                 # Next.js 15 App Router + Tailwind
    app/                    # 33 routes (24 route folders)
    components/ lib/        # UI components, hooks, API clients
    __tests__/              # 76 Vitest suites
  mobile/                   # Flutter 3.x (see the PRM393 paper)
    lib/{core,data,logic,presentation}/
    test/{data,logic,presentation}/
```

Table IX
ENDPOINT INVENTORY BY MODULE AND HTTP METHOD.

Module	Base path	GET	POST	PUT	PATCH	DELETE
programs/programs	/api/v1/programs	10	11	0	2	2
programs/live-studio	/api/v1/programs/studio	5	11	0	0	0
channels/channels	/api/v1/channels	7	4	0	1	1
users/users	/api/v1/users	5	1	0	4	0
auth/auth	/api/v1/auth	1	8	0	0	0
programs/ingest/ingest	/api/v1/programs/ingest	1	8	0	0	0
programs/social	/api/v1	4	4	0	0	1
notifications/notifications	/api/v1	2	3	0	0	2
ai-curator/ai-curator	/api/v1/ai-curator	2	4	0	0	0
audit-logger/audit-logger	/api/v1/audit-logs	6	0	0	0	0
watchlist/watchlist	/api/v1/me/watchlist	2	2	0	0	1
billing/billing	/api/v1/billing	2	2	0	0	0
cloudinary/cloudinary-upload	/api/v1/uploads/cloudinary	1	2	0	0	1
search/search	/api/v1/search	4	0	0	0	0
health/health	/api/v1/health	3	0	0	0	0
newsletter/newsletter	/api/v1/newsletter	1	2	0	0	0
programs/stream-health	/api/v1/programs/health	1	2	0	0	0
admin/admin	/api/v1/admin	1	1	0	0	0
programs/studio-catalog	/api/v1/programs/studio/catalog	2	0	0	0	0

Table X
SOURCE-CODE SCALE PER COMPONENT.

Component	Lines of code	Unit
Backend NestJS (src/)	27,826	file .ts
Backend test (test/)	4,512	file .ts
gRPC proto	83	file .proto
Audit-logger microservice	382	file .ts
Prisma schema	683	file .prisma
Frontend Next.js (app/, components/, lib/)	51,742	file .ts/.tsx
Mobile Flutter (lib/)	53,922	file .dart
Mobile test (test/)	2,529	file .dart

```
  playwright-tests/          # 8 browser end-to-end
    scenarios
  .github/workflows/ci.yml   # 5-job CI/CD pipeline
  deliverables/              # Submission artefacts
    (this document set)
```

Listing 2. Directory structure at a high level.